

User Input



Question:
What diseases
are included in
the picture?

GT:
Lung cancer.

Output

Potential mass or
nodule in lung.
Ans: Lung cancer.

Proprietary
MLLM
Coordinator

Iterative
CoT-Answer
Review

Step #1:
Find Entity

Response:
"Lung"

Thinking



Entity
Proposal VLM

Step #2:
Segmentation

Response:
Mask & C



Segmentation
Model

Step #3:
G-VQA

Response:
"Lung cancer"

Thinking



EG-VQA
VLM

(a) Medical Entity Proposal

Original Image

User
Question +

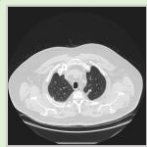
Please find the
most relevant
entity based on
user question.

RFT
InternVL-3

CoT: The lungs appear to
have some areas of
increased density which
could indicate the presence
of certain diseases...

Ans: Lung

(b) Entity Referring Segmentation



"Lung"

Image Proj.

BERT
Enc.

SAM
Layers

Text Proj.

SAM
Dec.



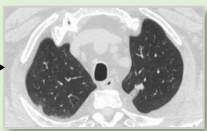
C = 95%

(c) Evidence-Grounded VQA

Evidence view

Zoom-in
Mask
Global

Original Image



User Question

SFT+RFT
InternVL-3

CoT: Notably there
is a denser area,
suggesting the
presence of a mass
or nodule...

Ans: Lung cancer